ICS 2315: Data Architecture & Warehousing - Assignment Guidelines

Reference Material

Core Textbook

- **Title:** Database System Concepts (7th Edition)
- **Chapters to Review:** 10, 11, 12, 13, 14, and 20

Reading Focus: Chapter 10 (Big Data)

- **Sections:** 10.0 through 10.3
- **Page 467 onwards:** Sources & uses of Big Data, IoT, Big Data Storage Systems, File-block-data Distributed file system, Sharding, and Types of Partitioning.
- **Page 483:** MapReduce Paradigm.

Coursework Assignments

Assignment	Weight	Core Focus	Key Tools & Databases	Dataset Source
Assignment 1	10 Marks	NoSQL Data Processing	Apache, Python, MongoDB (indexing focus), Cassandra	NSE (Nairobi Securities Exchange) Data
Assignment 2	10 Marks	Graph Data Analytics	Neo4j (via Neo4j.com)	Traffic Data
Final Deliverable	10 Marks	Summary Report	Comprehensive documentation of Assignment 1 & 2 tasks	N/A

Assignment Details

Assignment 1: NoSQL Data Processing (10 Marks)

- **Task:** Process and analyze large-scale structured/unstructured data using NoSQL storage solutions.
- **Required Tools:** Apache ecosystem, Python.
- **Database Systems:**
    - **MongoDB:** Implementation focus on indexing strategies and query optimization.
    - **Cassandra:** Distributed architecture setup and data modeling.
- **Dataset:** Nairobi Securities Exchange (NSE) Data *(To be distributed by the lecturer)*.

Assignment 2: Graph Data Analytics (10 Marks)

- **Task:** Build analytics models and visualize complex entities and their interactions using graph database structures.
- **Tool:** Neo4j (access via Neo4j.com).
- **Core Concepts:** Nodes (entities) and Links (relationships).
- **Dataset:** Traffic Data.
- **Implementation Details:**
    - Model **"Accidents"** and **"Traffic"** as core nodes.
    - Define and rate relationships based on severity parameters (e.g., classifying impact levels such as "severe").

Final Deliverable (10 Marks)

- **Submission Requirement:** Prepare and submit a comprehensive Summary Report that thoroughly documents the methodology, implementation code, schema designs, and analytical findings for both Assignment 1 and Assignment 2.